

Constantinos Paschalis



📍 Zurich, Switzerland
☎ +41 078 331 96 48
✉ Con.paschalis@gmail.com
🌐 linkedin.com/in/con-paschalis
🌐 Portfolio

Profile

MSc Energy Science and Technology student at ETH Zürich specializing in energy systems modeling and optimization. Experience in techno-economic analysis, heat pump systems, and CO₂ transport modeling across academic and industry settings.

Education

ETH Zürich , Zurich, Switzerland MSc Energy Science and Technology	2024 – Present
Texas A&M University , Doha, Qatar Energy and Sustainability Summer School	2024
University of Cyprus , Nicosia, Cyprus BSc Mechanical Engineering	2020 – 2024

Experience

EnergyIntel Services Ltd , Nicosia, Cyprus R&D Intern – Modeled thermal energy storage systems and evaluated fluid selection using multi-criteria analysis.	Jun 2025 – Aug 2025
University of Cyprus , Nicosia, Cyprus Department & School Representative – Represented students in faculty meetings, contributing to academic policy discussions and coordinating communication between students and administration.	Nov 2022 – Nov 2023
Texas A&M University , College Station, TX, USA Research Assistant (HVAC) – Performed hardware-in-the-loop and simulation-based analysis of HVAC systems under varying climate conditions.	Jun 2023 – Jul 2023

Projects

CO₂ Transport Network Resilience – ETH Zürich
Built techno-economic model of CO₂ transport resilience, showing resilience is cost-effective only under high utilization and prolonged failures, with temporary storage as the optimal strategy.

Mega Heat Pump Economic Evaluation – ETH Zürich & Everllence
Developed techno-economic model integrating thermodynamics and electricity markets, demonstrating revenue potential from FCR participation under varying market conditions.

Household Energy Optimization – ETH Zürich
Built MILP model and local app, enabling optimal appliance selection under dynamic pricing and CO₂ constraints.

Skills

Programming: Python, MATLAB
Modeling: ANSYS, SolidWorks
Core: Energy Systems, Thermodynamics, Optimization
Languages: Greek (Native), English (Fluent)

Awards

ETEK, AHK, Elysse Irrigation LTD (2024) – Top Mechanical Engineering Graduate (9.06/10)
MUSKITA Awards (2022, 2023) – Highest academic performance (2nd & 3rd year)